PulseCoach: Cardiac-Timed XR Stress Regulation Coaching with State-Aware AI

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Abstract

Emergency physicians handing off a critical patient, crisis negotiators managing an active standoff, and de-escalation officers facing an armed subject are trained on what to do under pressure, but rarely on whether their nervous system can sustain performance while under it. Composure under pressure is not a behavior but an autonomic state. A trainee can deliver every line correctly while their nervous system is still in alarm. Existing VR rehearsal systems cannot tell the difference. PulseCoach can.

Keywords

human-computer interaction, physiological computing, extended reality, stress regulation, cardiac-phase timing, AI coaching, EmotiBit

1 Introduction

Self-regulation under pressure is one of the hardest skills to train, and it surfaces wherever stakes are high. Whether the moment is an emergency physician's medical handoff, a tense conflict de-escalation, a crisis negotiation, or high-pressure public communication, a person can know exactly what to do and still lose access to it under stress. The skill people actually need is not memorizing answers or actions. It is the ability to notice their own physiological reaction, recover from it, and continue [7]. This skill is trainable. Studies on virtual reality exposure therapy show that immersive practice can produce measurable improvements in anxiety regulation when conditions are arranged correctly [2].

AI-driven training systems have begun to address this across many domains. Commercial platforms like VirtualSpeech and Mursion provide scripted practice scenarios with feedback on verbal behavior, eye contact, and vocal tone, while research systems like MACH extend this with conversational AI agents that adapt scenarios and rate nonverbal behavior [9, 10]. Mixed reality (MR) extends this approach further by placing the trainee in an immersive simulated scenario, eliciting genuine stress responses through presence and scenario realism [17]. Recent work has shown that embodied AI agents in MR environments outperform voice-only agents in collaborative learning tasks [18]. However, these systems share a critical limitation: they evaluate what the trainee says and does, but cannot read whether the trainee is genuinely composed or simply passing surface metrics. They cannot certify readiness, adapt pressure to the individual, or deliver interventions at the moment the body is most able to use them.

AI-supported interview training has explored this space extensively. Systems like SimInterview use large language models to generate adaptive, personalized scenarios [12], while VR-based platforms simulate realistic communication pressure through embodied presence [5]. These systems show that conversational AI can meaningfully scaffold high-stakes communication practice. However, they share the same ceiling as behavioral systems: they measure what the trainee says and does, not whether their nervous system is regulated. Self-regulation under pressure is not a communication skill, it is an autonomic one. For a physician handing off a critical patient or a negotiator managing an escalating standoff, the question is not whether they know what to say. It is whether their body can access it.

2 System Design

PulseCoach supports high-stakes communication scenarios where composure under pressure determines outcomes, including emergency physician medical handoffs, conflict de-escalation, and crisis negotiation. The system does not certify what the trainee says or does. It certifies that the trainee's body learned to regulate. PulseCoach achieves this through three coordinated modules: a physiological sensing module that reads the trainee's autonomic state in real time, a biometric-adaptive scenario engine that tunes scenario pressure to that state, and a state-aware AI coach that delivers cardiac-phase-timed interventions through XREAL glasses.

2.1 Physiological Sensing Module

The physiological sensing module is built around EmotiBit, a research-grade wearable worn on the wrist. EmotiBit streams four channels simultaneously into the XR pipeline over WiFi using the Lab Streaming Layer (LSL) protocol [11]: electrodermal activity for stimulus-locked arousal detection [6], heart rate variability for vagal recovery analysis [4, 16], per-beat photoplethysmography for cardiac-phase scheduling [1], and 9-axis IMU motion data for posture, fidget detection, and motion-artifact gating of the PPG stream [3].

These signals serve three roles. First, they enable the scenario engine to evaluate the trainee's actual physiological state rather than their self-report, gating scenario escalation on confirmed vagal recovery rather than performed calm. Second, they feed the coaching agent's context window, so its responses are shaped by the trainee's live state rather than their verbal input alone. Third, the per-beat PPG stream enables cardiac-phase scheduling: R-peak detection with millisecond timestamp precision allows the system to predict the next diastolic window 300–500 ms ahead and deliver AR interventions at the moment the body is most receptive to calming input [8, 13].

Consumer wearables such as Apple Watch, Fitbit, and Garmin expose only aggregated outputs designed for fitness tracking, not the raw real-time streams required for closed-loop physiological

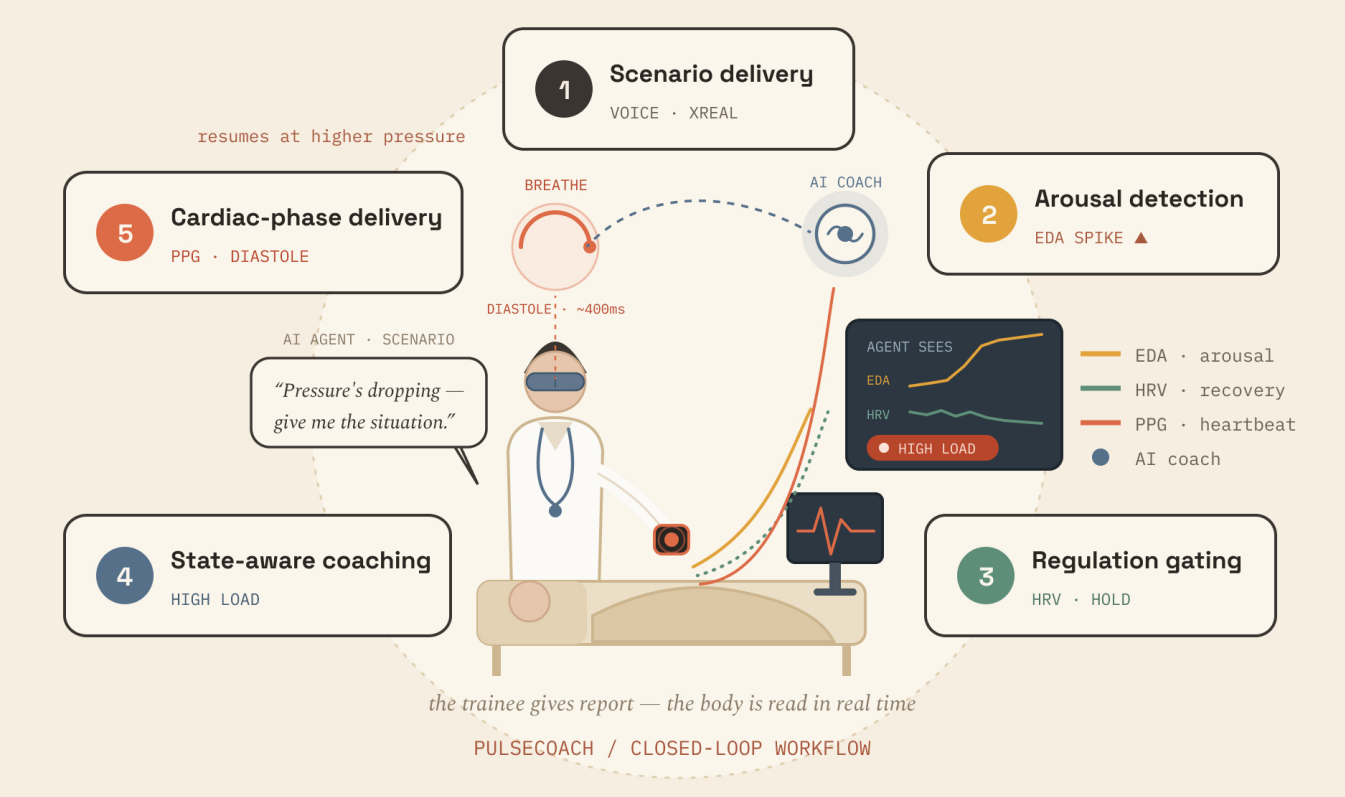


Figure 1: PulseCoach closed-loop pipeline: EmotiBit reads autonomic state in real time across five stages: scenario delivery, arousal detection, regulation gating, state-aware coaching, and cardiac-phase-timed AR delivery at the diastolic window through XREAL glasses.

computing. EmotiBit is the only research-grade wearable that exposes all four signal types as raw real-time streams suitable for this pipeline.

2.2 State-Aware AI Coaching Module

The state-aware AI coach decides what intervention to deliver and when the trainee needs one. It receives a context window containing two streams: the scenario transcript and a live biometric state summary updated every 5 seconds (EDA level, HRV recovery status, recent spike count, and posture state from the IMU). This biometric context shapes how the agent responds, operationalizing state-awareness at the agent level rather than in a separate rules-based system [14, 15].

When the agent detects HIGH LOAD (elevated EDA combined with suppressed HRV), it generates a coaching intervention appropriate to the moment: a paced breathing cue, an encouraging redirect, or a pause prompt. The agent then hands the intervention to the MR module for cardiac-phase-timed delivery.

2.3 MR Module

The MR module delivers the AI coach's interventions through XREAL glasses at moments timed to the cardiac cycle. The cardiac-phase scheduler uses the per-beat PPG stream to predict the next diastolic window 300–500 ms ahead. When the AI coach hands off an intervention, the scheduler queues it for delivery into that window, placing the cue when the body is most able to use it [8, 13]. The AR overlay renderer then displays the intervention as an ambient cue in the trainee's peripheral vision: a paced breathing arc, a brief text cue, or a soft visual signal, without interrupting the scenario or breaking immersion.

3 Technical Implementation

The pipeline is implemented in Unity 6, with XREAL SDK handling AR rendering. EmotiBit streams over WiFi to a laptop running an LSL inlet, and a custom Unity-LSL bridge synchronizes biometric timestamps with the scenario event log at millisecond precision. The cardiac-phase scheduler runs in its own thread, maintaining a rolling 2-second buffer of detected R-peaks to estimate heart rate and predict the next diastolic window. The AR overlay renderer subscribes to this scheduler and queues the breathing-guide asset for delivery at the predicted moment. IMU data is processed in parallel, gating the PPG stream during motion artifacts to keep R-peak detection clean. Professor Jin's lab has working Unity 6 and XREAL SDK pipelines from prior XR learning research [18], and EmotiBit ships with documented LSL streaming; the primary build work is the cardiac-phase scheduler and the Unity-LSL bridge, both scoped to fit the contest timeline.

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